
Quantile-Initialised Ordinal Collaborative Filtering with Gated Auxiliary Fusion

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Abstract

We propose a simple calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the empirical rating quantiles of the training set, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. The recipe matches the data marginal exactly at the start of training and, on a transparent dot-product matrix-factorisation backbone, achieves low test negative log-likelihood (1.2584 across the five canonical MovieLens-100K splits, vs the $\log 5 \approx 1.61$ uniform baseline) without sacrificing RMSE. Paired with per-dimension gated fusion of side information, the gated+ordinal configuration attains RMSE 0.9124 ± 0.0036 across the five splits, beating both a no-fusion MF analogue (0.9160) and additive fusion (0.9189) on the across-split mean. The contribution is practical rather than algorithmic: standard components are combined into a reproducible, calibrated recommender pipeline.

1 Introduction

Collaborative filtering (CF) for explicit feedback is commonly framed as predicting an unobserved scalar rating r_{ui} from a sparse user-item rating matrix $R \in \mathbb{R}^{n \times m}$. Latent-factor models based on matrix factorisation [3] remain standard baselines and building blocks on MovieLens [2]. Two persistent issues remain when these models are applied as a course-style data-analytics tool rather than a production recommender. First, side information (user demographics, item genres) is typically injected by adding a projected feature vector to the latent embedding, with no learned mechanism to decide *when* to trust the embedding versus the side information. Second, the rating is treated as a continuous quantity even though it is in fact a five-class ordinal variable; the resulting predictions can be RMSE-competitive yet badly calibrated as distributions over star ratings.

Related work. Gated content fusion has been explored by GATE [5]; ordinal regression starts from McCullagh’s cumulative-link model [6] and has been applied to MovieLens by OrdRec [4]. Recent ordinal recommender baselines include OrdNMF [1] and OPRFM [7]. Our contribution is not a new family of models, but a compact integration of gated side information and calibrated cumulative-link prediction on a transparent MF core.

Contributions. We make two practical contributions. First, we initialise cumulative-link thresholds from the training-set rating quantiles and exclude those thresholds from weight decay, so the ordinal head starts with the correct rating marginal. Second, we pair that head with symmetric per-dimension gates that blend ID embeddings with user and item side-feature projections. The ablation separates their roles: gating gives the larger RMSE change, while the ordinal head gives calibrated class probabilities and better boundary-class behaviour.

2 Problem Definition

Let $\mathcal{U} = \{1, \dots, n\}$ be the set of users and $\mathcal{I} = \{1, \dots, m\}$ the set of items. Observed ratings form a sparse set $\mathcal{D} = \{(u_t, i_t, r_t)\}_{t=1}^N$ with $r_t \in \{1, 2, 3, 4, 5\}$. Each user u carries a side-information vector $f_u \in \mathbb{R}^{d_u}$ (z-scored age, one-hot gender, one-hot occupation in MovieLens-100K, $d_u = 24$) and each item i carries $g_i \in \mathbb{R}^{d_i}$ (multi-hot genre indicator, $d_i = 19$). The task is to predict $\hat{r}_{ui}$ for held-out (u, i) pairs, evaluated by RMSE, MAE, classification accuracy of the rounded prediction, and where applicable the held-out negative log-likelihood (NLL) of the predicted rating distribution.

3 Method

3.1 Matrix factorisation backbone

We start from a biased dot-product MF model. Each user and item carries a d -dimensional embedding ($p_u, q_i \in \mathbb{R}^d$) and a scalar bias ($b_u, b_i \in \mathbb{R}$). The pre-head score is

$$s_{ui} = p'_u \cdot q'_i + b_u + b_i, \quad (1)$$

where p'_u, q'_i are the (possibly fused) representations defined below. With fusion disabled, $p'_u \equiv p_u$ and $q'_i \equiv q_i$.

3.2 Gated auxiliary fusion

Let $a_u = \text{ReLU}(W_u f_u + b_{f,u})$ be a learned projection of the user side features into the embedding space, and similarly $a_i = \text{ReLU}(W_i g_i + b_{f,i})$ for items. We define the fused user representation as

$$g_u = \sigma(W_g [p_u \parallel a_u] + b_g), \quad p'_u = (\mathbf{1} - g_u) \odot p_u + g_u \odot a_u, \quad (2)$$

where $\parallel$ denotes concatenation and $\odot$ is the Hadamard product, so the gate $g_u \in [0, 1]^d$ acts *per dimension*. The item-side fusion is identical with (q_i, a_i) in place of (p_u, a_u) . We zero-initialise both W_g and b_g , which makes $g_u \equiv \frac{1}{2}\mathbf{1}$ at training start. This gives every dimension the same neutral starting mixture between the ID embedding and the ReLU-projected side features; the gate then opens and closes per dimension as gradient flows. The ablation compares this against *additive* fusion ($p'_u = p_u + a_u$) and *none* fusion ($p'_u \equiv p_u$), with analogous item-side definitions.

3.3 Ordinal regression head

For the head we treat r as an ordinal variable with $K = 5$ classes. We use the cumulative-link model of McCullagh [6] with monotone thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$:

$$P(r \leq k \mid s) = \sigma(\theta_k - s), \quad P(r = k \mid s) = \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s), \quad (3)$$

with $\theta_0 = -\infty$ and $\theta_5 = +\infty$. We enforce monotonicity by the softplus reparameterisation

$$\theta_k = \theta_1 + \sum_{j=1}^{k-1} \text{softplus}(\delta_j), \quad (4)$$

which guarantees $\theta_k < \theta_{k+1}$ for any unconstrained parameters.

Empirical-quantile initialisation. We initialise from the empirical rating distribution. Let $\hat{p}_k$ be the training-set frequency of rating k and $\hat{F}_k = \sum_{j \leq k} \hat{p}_j$ its cumulative. Choosing

$$\theta_k = \text{logit}(\hat{F}_k), \quad \delta_j = \log(\exp(\theta_{j+1} - \theta_j) - 1) \quad (5)$$

guarantees $P(r \leq k \mid s=0) = \hat{F}_k$, so the predicted marginal at the start of training equals the data marginal exactly. The second equation is just the softplus inverse, recovering the unconstrained δ_j that the optimiser then updates. We also exclude $\{\theta_1, \delta_j\}$ from weight decay because their scale carries calibration information.

The training loss is the negative log-likelihood $\mathcal{L}_{\text{ord}} = -\frac{1}{N} \sum_t \log P(r = r_t \mid s_{u_t i_t})$, and the point prediction is the expected rating $\hat{r}_{ui} = \sum_{k=1}^5 k P(r = k \mid s_{ui})$. As an ablation we also consider the standard *sigmoid* head: $\hat{r}_{ui} = 4\sigma(s_{ui}) + 1$ trained with mean-squared error.

Algorithm 1 Forward pass + ordinal NLL for one observed rating (u, i, r) .

- 1: $a_u \leftarrow \text{ReLU}(W_u f_u + b_{f,u}); a_i \leftarrow \text{ReLU}(W_i g_i + b_{f,i})$ $\triangleright$ side projections
 - 2: $g_u \leftarrow \sigma(W_g^{(u)}[E_u[u] \| a_u] + b_g^{(u)}); g_i \leftarrow \sigma(W_g^{(i)}[E_i[i] \| a_i] + b_g^{(i)})$ $\triangleright$ per-dim gates
 - 3: $p' \leftarrow (\mathbf{1} - g_u) \odot E_u[u] + g_u \odot a_u; q' \leftarrow (\mathbf{1} - g_i) \odot E_i[i] + g_i \odot a_i$
 - 4: $s \leftarrow p' \cdot q' + b_u + b_i; \theta_k \leftarrow \theta_1 + \sum_{j < k} \text{softplus}(\delta_j)$
 - 5: $P(r=k \mid s) \leftarrow \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s)$, with $\theta_0 = -\infty, \theta_K = \infty$
 - 6: **return** $\mathcal{L} = -\log P(r=r_{\text{true}} \mid s)$
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3.4 Complexity

The MF backbone has $(n + m)d$ embedding parameters plus $n + m$ biases; for MovieLens-100K with $n = 943, m = 1682, d = 128$, that is $\approx 3.4 \times 10^5$ parameters. Each fusion module adds a projection ($d_{u/i} \cdot d + d$) and a gate ($2d^2 + d$); summed over both sides this is $\approx 7 \times 10^4$ parameters — about one-fifth of the embedding budget. The ordinal head adds only $K - 1 = 4$ parameters, and the dominant per-rating cost remains $O(d)$.

4 Experiments

4.1 Setup

We use MovieLens-100K [2] on its five canonical 80/20 splits u1–u5, each yielding 80,000 training and 20,000 test ratings on the same 943 users and 1,682 items. Headline numbers are reported as mean $\pm$ std across the five per-split means (each split-mean itself averages three training seeds), giving 15 trained models per ablation cell. User side features are z-scored age, two-class one-hot gender, and 21-class one-hot occupation ($d_u = 24$); item features are 19-genre multi-hot ($d_i = 19$). All experiments use Adam with learning rate 10^{-3} , $d = 128$, and weight decay $\lambda = 10^{-5}$ on all parameters except the ordinal-head thresholds. Early stopping is driven by validation RMSE only, and the test set is read once after restoring the best-validation weights.

4.2 Baselines and metrics

The ablation table includes the no-fusion sigmoid model as a biased-MF analogue, additive fusion, and gated fusion, each under sigmoid and ordinal heads. Metrics are RMSE, MAE, rounded-rating accuracy, and NLL where a native class-probability head is available.

4.3 Results

As a sanity check, the proposed model also beats three transparent baselines on the u1 split under the headline protocol (single seed, patience 30), shown in Table 1. The multi-split ablation below is the main evidence, but this table guards against only comparing the model to its own variants.

Table 1: Transparent baselines on MovieLens-100K u1. Lower is better for RMSE/MAE/NLL; higher is better for accuracy.

Model	RMSE	MAE	Accuracy	NLL
SVD (rank 20)	1.0834	0.8896	0.3716	—
Biased MF	0.9221	0.7274	0.4124	—
NMF	0.9197	0.7252	0.4146	—
Gated+ordinal	0.9122	0.7130	0.4367	1.2472

The ablation matrix in Table 2 reports the full $\{\text{none, additive, gated}\} \times \{\text{sigmoid, ordinal}\}$ cross-product on three seeds. The proposed configuration — gated fusion with the ordinal head — has the best mean value on all four reported metrics. Across the five splits it improves RMSE by 0.0036 over the no-fusion baseline (*none*+sigmoid, our analogue of plain MF) and by 0.0065 over the additive-fusion variant under the sigmoid head. Replacing only the head from sigmoid to ordinal

on the same gated backbone gives $\Delta = +0.0011$ RMSE (within between-split std), and improves MAE by 0.0056 and accuracy by 0.0087. The ordinal cell is also the only one that emits a native class distribution (NLL 1.2584). Because the RMSE differences are small relative to between-split variation, we interpret the main value of the ordinal head as probabilistic output and per-class behaviour rather than a large RMSE improvement.

Table 2: Ablation matrix on MovieLens-100K, mean $\pm$ std across the five canonical splits u1–u5 (each split-mean averages three seeds). Bold = best per column. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.9160 $\pm$ 0.0061	0.7241 $\pm$ 0.0059	0.4233 $\pm$ 0.0044	—
none	ordinal	0.9170 $\pm$ 0.0072	0.7204 $\pm$ 0.0069	0.4295 $\pm$ 0.0056	1.2634 $\pm$ 0.0050
additive	sigmoid	0.9189 $\pm$ 0.0044	0.7238 $\pm$ 0.0055	0.4243 $\pm$ 0.0057	—
additive	ordinal	0.9174 $\pm$ 0.0040	0.7173 $\pm$ 0.0035	0.4340 $\pm$ 0.0038	1.2654 $\pm$ 0.0017
gated	sigmoid	0.9135 $\pm$ 0.0037	0.7182 $\pm$ 0.0034	0.4286 $\pm$ 0.0054	—
gated	ordinal	0.9124 $\pm$ 0.0036	0.7126 $\pm$ 0.0042	0.4373 $\pm$ 0.0045	1.2584 $\pm$ 0.0043

Interpretation. Table 2 supports two claims. First, gated fusion is the main RMSE mechanism: holding the head fixed, replacing *none* with *gated* drops RMSE by 0.0025 (sigmoid) or 0.0046 (ordinal), while replacing *additive* with *gated* drops it by about 0.005 under either head. Second, the ordinal head is mainly a probability/calibration improvement. Switching sigmoid to ordinal changes RMSE by at most 0.0015, but consistently improves MAE and accuracy and is the only head with native NLL. The proposed configuration’s NLL 1.2584 is well below the $\log 5 \approx 1.609$ uniform baseline.

Class-level behaviour. On `u1.test`, the ordinal head improves macro-F1 from 0.341 to 0.369 on the same gated backbone, mostly by recovering boundary ratings (class-1 recall 0.169 vs 0.102, class-5 recall 0.235 vs 0.204). Per-class ECE shows the same pattern: the ordinal head is better on classes 1, 2, 3, and 5, while the kernelised sigmoid head is better on the dominant class 4. This is why we treat the ordinal head as a probability/class-behaviour improvement rather than a large RMSE improvement.



Figure 1: Per-class F1 on `u1.test`. The ordinal head mainly improves the boundary classes, rather than creating a uniform gain across all ratings.

The calibration view is consistent with the class-level result: pooled ECE is similar for the kernelised sigmoid and ordinal heads, but the ordinal head is better on classes 1, 2, 3, and 5, while the sigmoid head wins only on the dominant class 4.

Figure 3 shows the sensitivity sweep over embedding dimension d and weight decay λ . The chosen $d = 128$ is the elbow: $d = 256$ improves RMSE by only 0.0024 at 34% extra wall time, comparable to seed-to-seed variance. The best tested regularisation is $\lambda = 10^{-5}$; $\lambda = 10^{-4}$ overregularises by 0.0080 RMSE.

Embedding and regime diagnostics. The post-fusion embeddings contain side-information structure without an auxiliary classification loss. In 128-D, the silhouette score is +0.0848 for

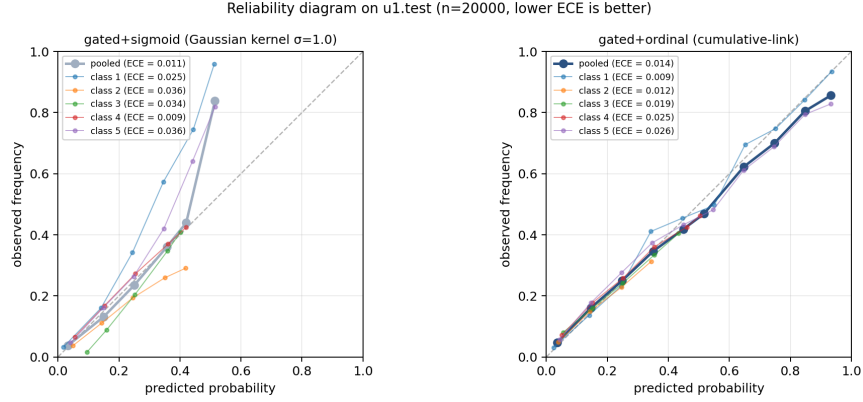


Figure 2: Reliability diagrams on `u1.test`. The ordinal head spreads probability mass more evenly across rating classes, while the sigmoid head is strongest on the dominant class.

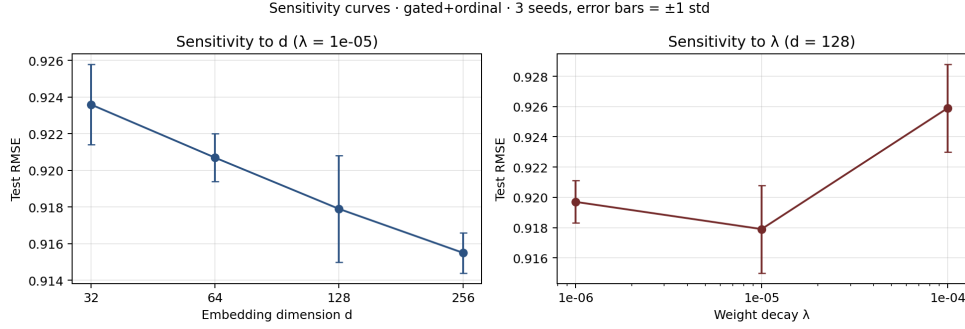


Figure 3: Sensitivity of gated+ordinal test RMSE to embedding dimension d and weight decay λ on `u1`; error bars are ± 1 std over three seeds.

user occupation and +0.1930 for item dominant genre. After 2-D projection, item genre remains visible under IsoMap (+0.232) but is much weaker under PCA (+0.048), suggesting that the item-side genre geometry is curved. The user-side occupation silhouette becomes negative under all 2-D projections, so those features appear distributed across too many dimensions to visualise faithfully in a plane. This diagnostic is secondary to the RMSE/NLL result, but it checks that the gate is not merely adding opaque parameters.

Table 3: Cold-start RMSE by user activity. Negative Δ means additive fusion wins.

$ R_u $ bucket	< 30	30–59	60–119	120–239	≥ 240
gated RMSE	0.9799	0.9649	0.8841	0.8896	0.9200
additive RMSE	0.9803	0.9768	0.8929	0.8946	0.9128
Δ	+0.0004	+0.0119	+0.0088	+0.0051	−0.0071
mean g_u	0.246	0.256	0.263	0.268	0.291

The cold-start breakdown in Table 3 shows that gated fusion is not uniformly beneficial: it is strongest for mid-activity users, ties to the coldest bucket, and reverses for the 14 most active users. A three-cell MovieLens-1M check preserved the same ordering at larger scale, with gated+ordinal reaching RMSE 0.8481 and NLL 1.1842.

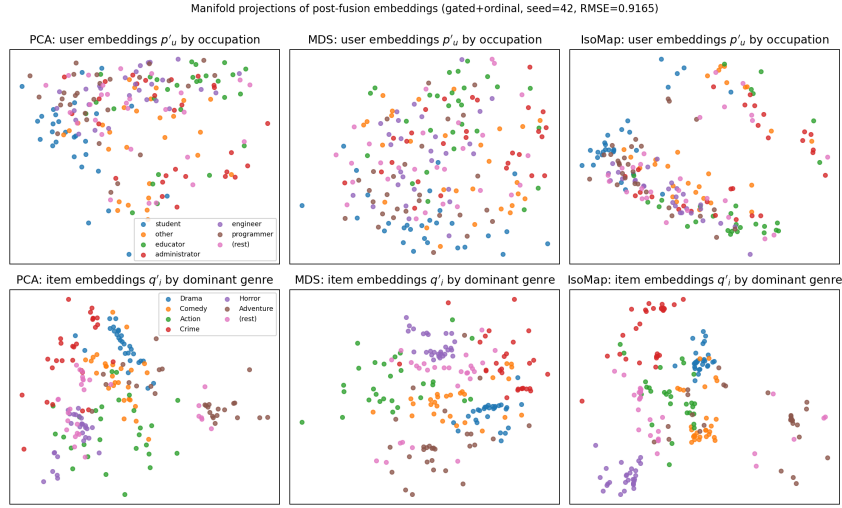


Figure 4: Two-dimensional projections of post-fusion user embeddings (top, occupation) and item embeddings (bottom, dominant genre). Columns show PCA, classical MDS, and IsoMap; the item-side genre pattern is clearest under the nonlinear projection.

Table 4: Scale-up to MovieLens-1M, random 80/10/10 split, mean $\pm$ std over three seeds.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.8567 ± 0.0008	0.6782 ± 0.0007	0.4480 ± 0.0005	—
gated	sigmoid	0.8502 ± 0.0003	0.6695 ± 0.0001	0.4579 ± 0.0002	—
gated	ordinal	0.8481 ± 0.0006	0.6630 ± 0.0010	0.4654 ± 0.0014	1.1842 ± 0.0013

4.4 Reproducibility

The code path is intentionally small enough to audit. All tables and figures in the report are regenerated from archived JSON/CSV outputs or from the scripts in Table 5. Each script records the model configuration, split, seed, and metric outputs used in the paper. The reported ablation uses training seeds {42, 43, 44} on each canonical split and fixes the validation-split seed to avoid mixing split variance with model-initialisation variance.

Table 5: Core reproduction scripts. Wall times are approximate on M1 Pro CPU.

Result	Script	Time
Main ablation, 6 cells, 5 splits	run_multisplit.py	~60 min
Baseline sanity check	run_baselines.py	~5 min
Sensitivity curves	run_sensitivity.py	~15 min
Class F1 and reliability	run_classmetrics.py, run_reliability.py	~4 min
Cold-start buckets	run_coldstart.py	~2 min
Embedding projections	run_visualization.py	~1 min
MovieLens-1M scale-up	run_mllm.py	~85 min

4.5 Discussion

The ablation paints a sharper picture than the headline number alone. Gated fusion accounts for most of the empirical RMSE change; the ordinal head provides native probability estimates cheaply — four learnable thresholds — but does not help RMSE on its own. This interaction suggests that ordinal heads are most useful here when paired with informative side-information mechanisms. Limitations are the small benchmark, coarse demographic/genre side features, and a rating-marginal calibration recipe that does not model per-user heteroscedasticity.

5 Conclusion

We presented a calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the training-set rating quantiles, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. Paired with per-dimension gated side information, the resulting model attains RMSE 0.9124 ± 0.0036 and NLL 1.2584 across the five MovieLens-100K splits, improving the across-split mean over both a no-fusion MF analogue and additive fusion. The main lesson is that gating drives the RMSE gain, while the ordinal head gives calibrated probabilities and better boundary-class behaviour at negligible parameter cost.

Practically, this means the ordinal layer should be read as a calibration interface rather than as the sole source of ranking improvement. It is useful when the backbone already has enough signal to separate rating boundaries, and it provides NLL and class probabilities that a sigmoid-MSE head cannot provide natively. The gated side-information path is the larger accuracy mechanism, but the cold-start breakdown shows that its gain is not uniform across activity regimes. A production extension would therefore keep the calibrated ordinal head, but allow the fusion strategy to vary by user activity instead of using one global gate behaviour everywhere.

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